

Introduction to Animation

12



Hi Thomas, what are you watching?



An animated movie. Have you watched any animated movie? All toys in it can walk and talk like humans do!



But toys cannot walk and talk. Toys are non-living things.



Yes. But this is animation. It is used to show objects move and talk like they are alive.



Can you show me how to do that?



Animation is a way to add movement to drawings, models and non-living objects. This makes them appear to move as if they are alive. All our favourite cartoon shows, and movies like Toy Story are animated. We can **animate objects and drawings** on computer. We can use animation software to do this. One such software is **Hue Animation**.

Introduction to Hue Animation

Hue Animation software helps us to create animation using still images. It uses a camera and a software to make the animation.

12. Introduction to Animation

A camera:

We use this to take pictures for our animation. It is called a Hue Camera and is different from our regular cameras. We will learn about it soon.

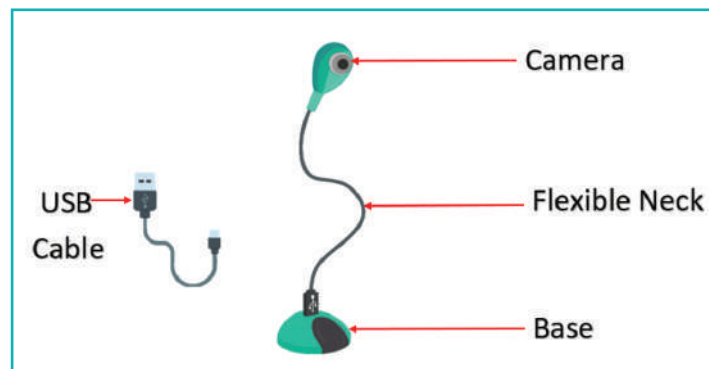
A software:

This software connects the camera and the computer. We use it to put images together.



Hue Animation

A Hue Camera is made up of different parts.



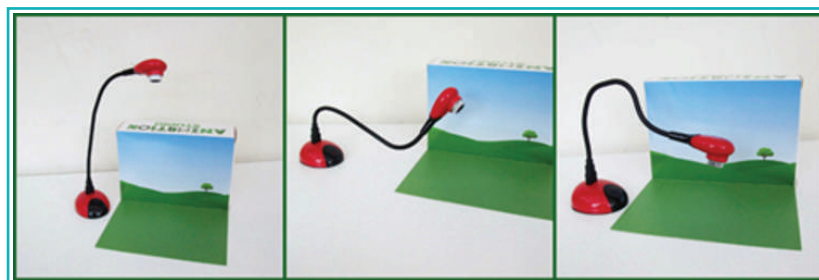
Hue camera

Camera:

This is used to capture images.

Flexible Neck:

The neck of the Hue Camera can be bent to move in any direction. We can use this to take pictures from different angles.



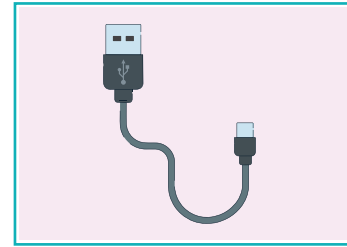
Flexible Neck

Healthy Practices:

Do not put pressure or forcefully bend the flexible neck

USB Cable:

A USB cable is used to connect the camera to the computer. It is plugged into the CPU of the computer.



USB cable

Exercise

Tick mark ☒ the correct answer.

- 1 A device used to capture images is called as
 - a. Flexible Neck ☐
 - b. Base ☐
 - c. Camera ☐
 - d. USB Cable ☐
- 2 Which part of the Hue Camera is useful to take pictures from different angles?
 - a. Camera ☐
 - b. Flexible Neck ☐
 - c. Images ☐
 - d. Computer ☐

Create an animation using Hue Animation Kit

In our animation, a pencil and eraser are meeting for the first time. They introduce themselves. To create this, we will need these things:

1. An eraser
2. A pencil
3. Two blank papers. One for the background and the other to make speech bubbles.

Follow these steps to create an animation. We will call it “Who am I?”

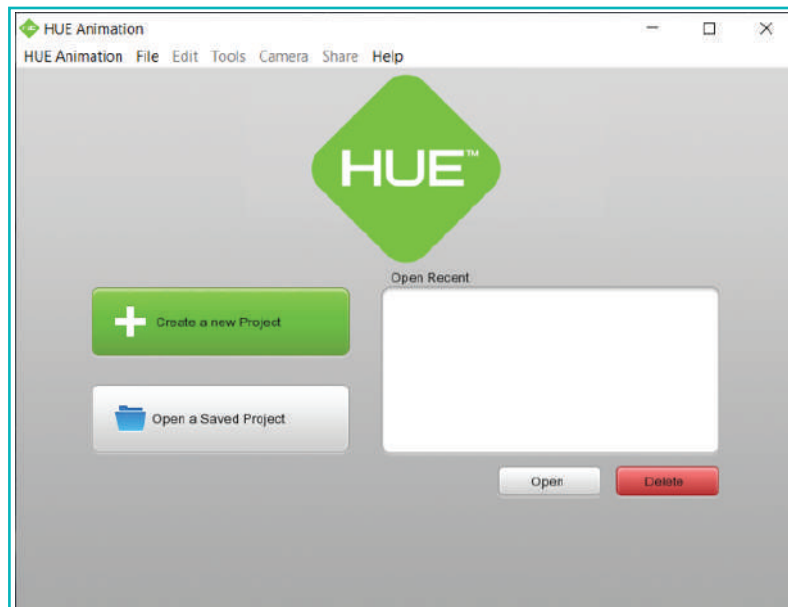
Steps

- 1 Open Hue Animation
 - Find the “Hue Animation” icon on your desktop.
 - Double click on it.



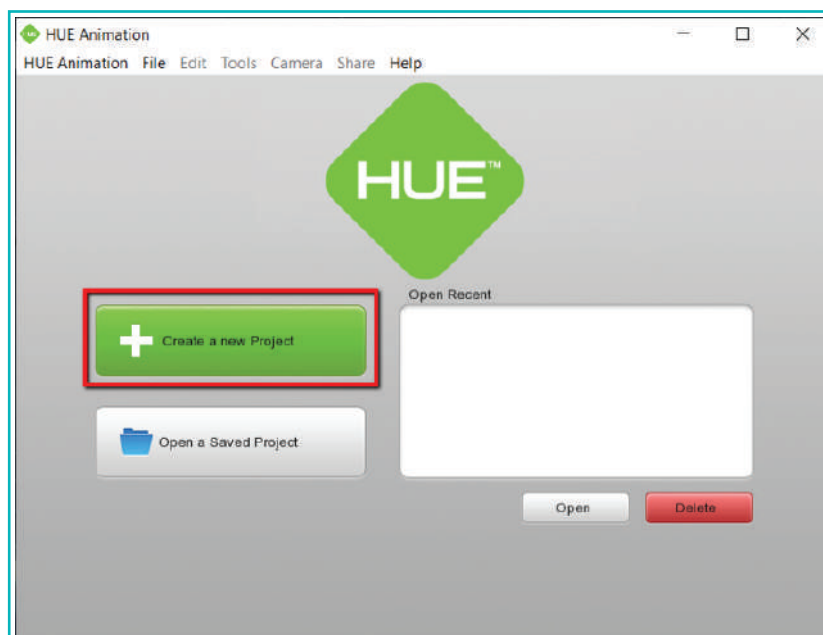
Steps

- Hue Animation software interface looks like this:



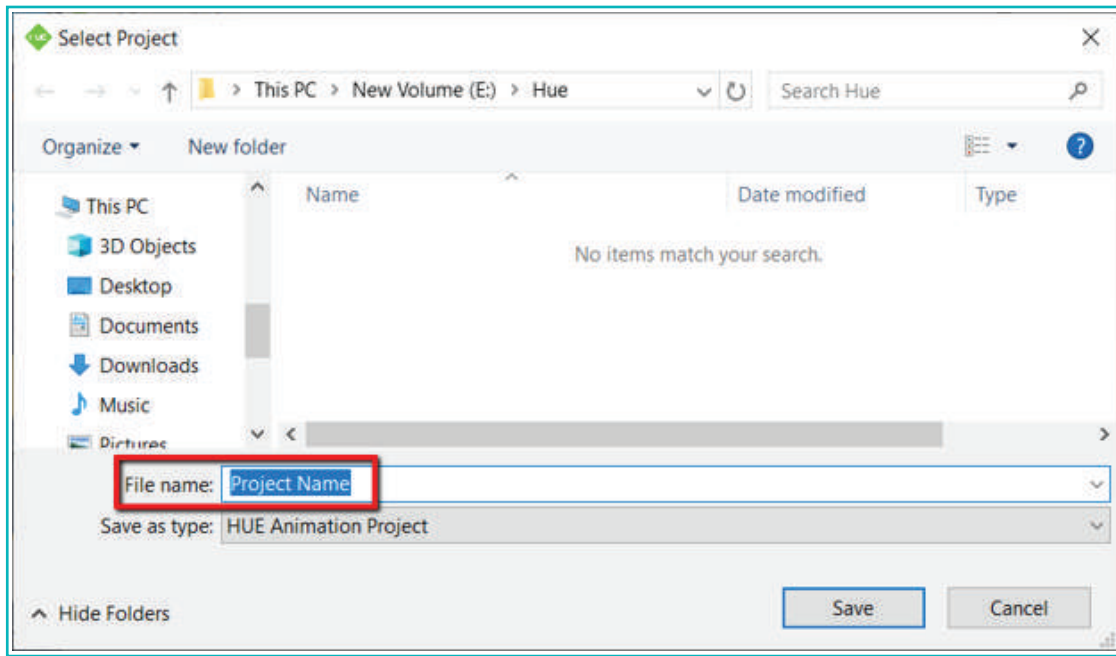
2 Create a new file/project

- To make an animation, we must create a new file.
- In the Hue Animation software, every new file is called a Project file.
- Click on “Create a new Project” to create a new file.



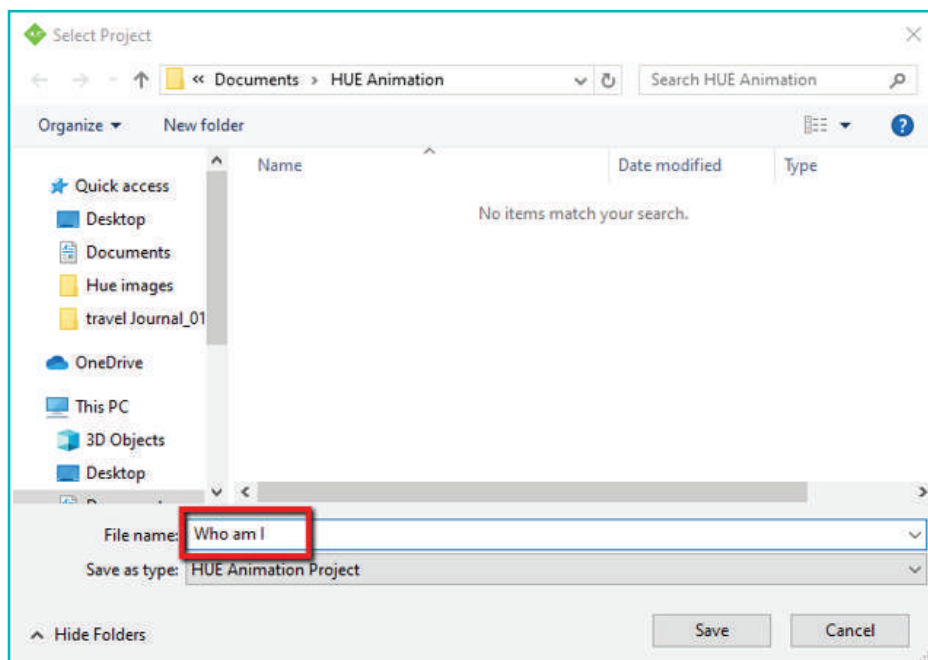
Steps

- Give a suitable name to the project.



Save project

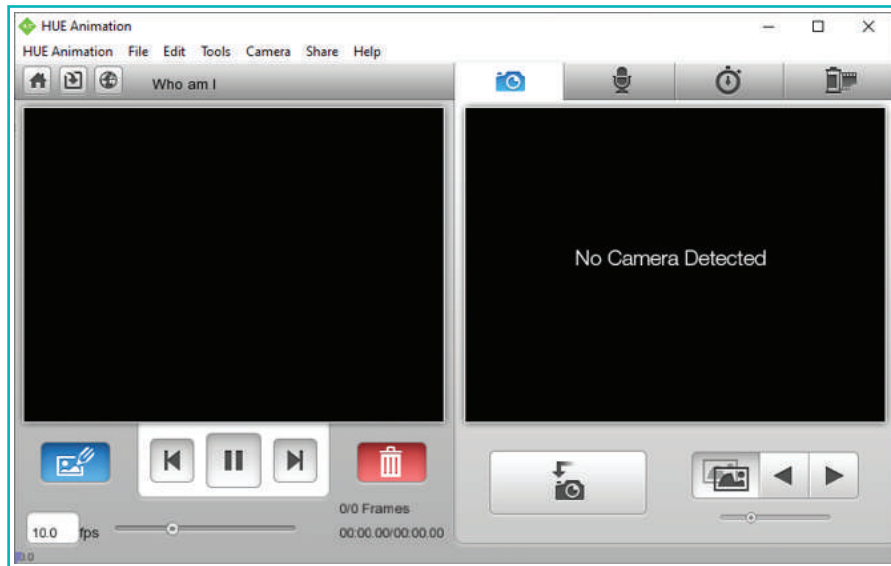
- Here, the name of animation is “Who am I?”. We will name the project “Who am I”.
- Click on the “Save” button to save the file.



Change the File name

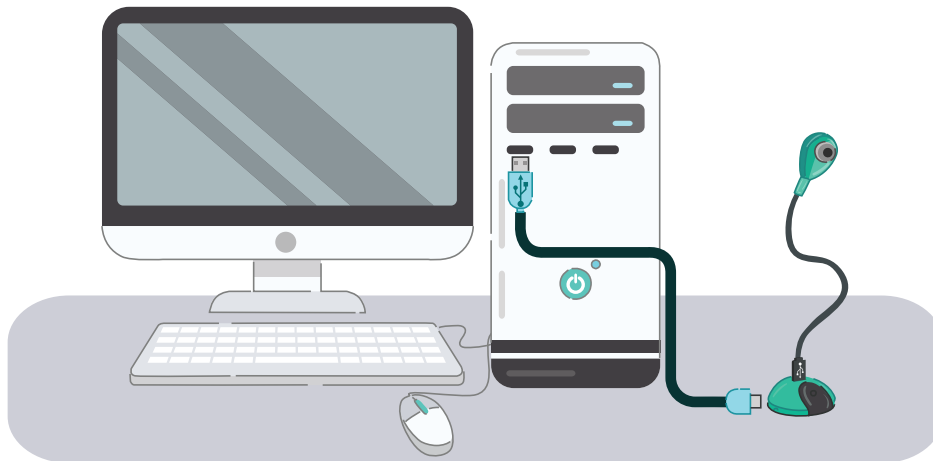
Steps

- 3 Set up camera
 - After we save the project, a **new window** opens. It is used to capture images for animation.



Capture Window

- Connect the camera to the computer with the **USB cable**.



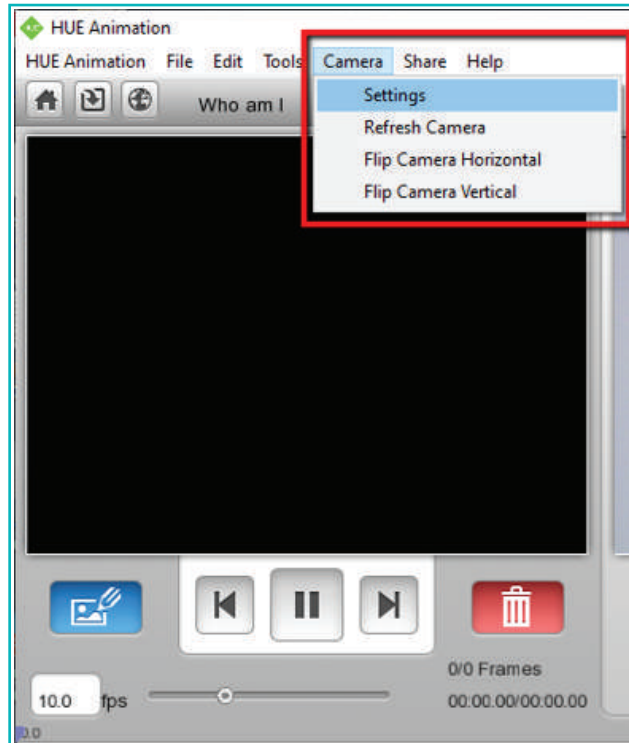
Connecting camera to computer

Healthy Practices:

Always connect the camera with the PC using the camera base, it will provide stability while taking pictures

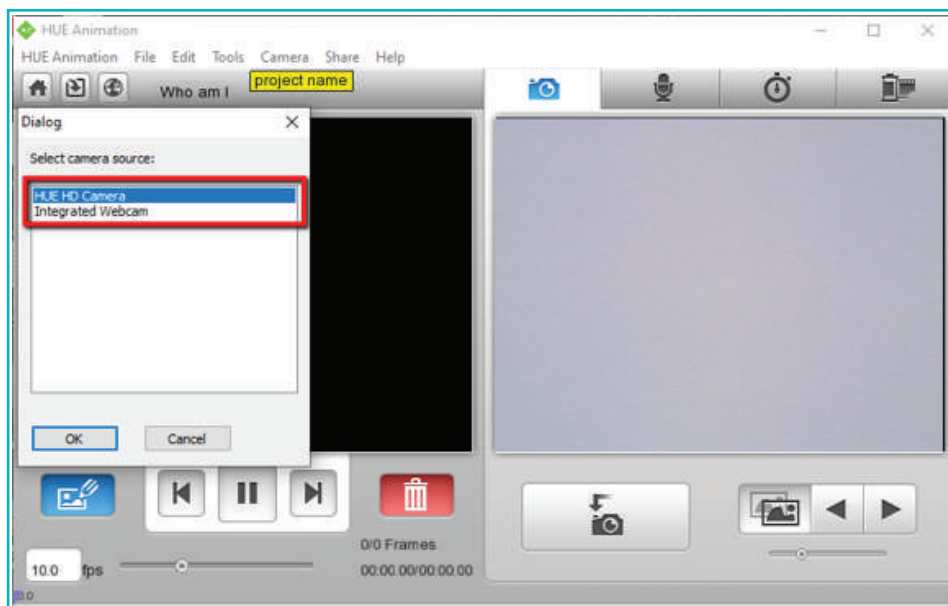
Steps

- In the Hue Animation window, open **Camera** and select **Settings**.



Camera settings

- Select **HUE HD Camera** and click on **OK**. It opens the Hue Camera.

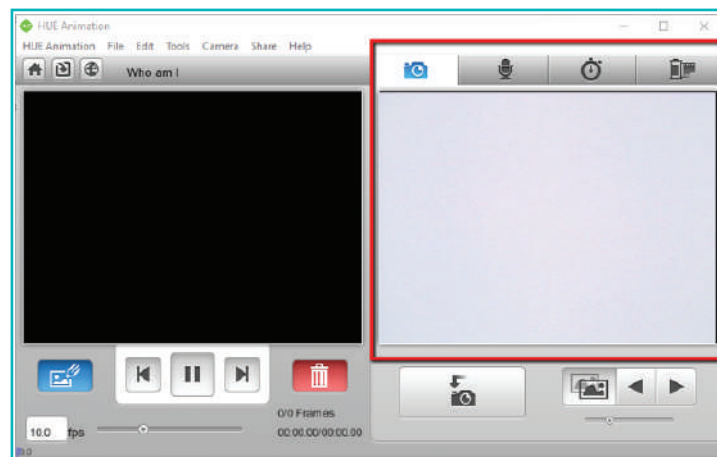


Selecting Hue HD camera

Steps

4 Capture Images and set up background

- First, create the background for the animation. Here, the background is white paper.
- Place a blank white paper in front of the camera.
- Make sure that you can see it well on the computer screen. This is our frame.
- The whole window must look white, like the image below.



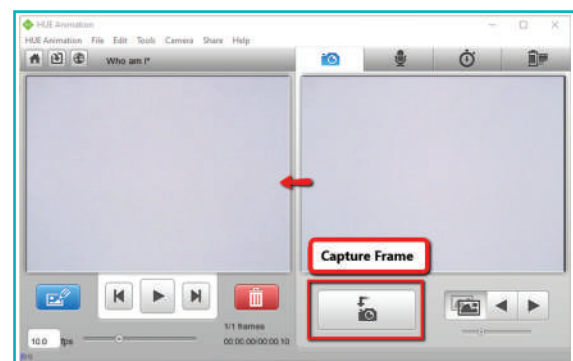
Place a white paper

Healthy Practices:

Do not damage or draw on the green platform and the image background on the box



- Click on the **Capture Frame** button to capture the image.

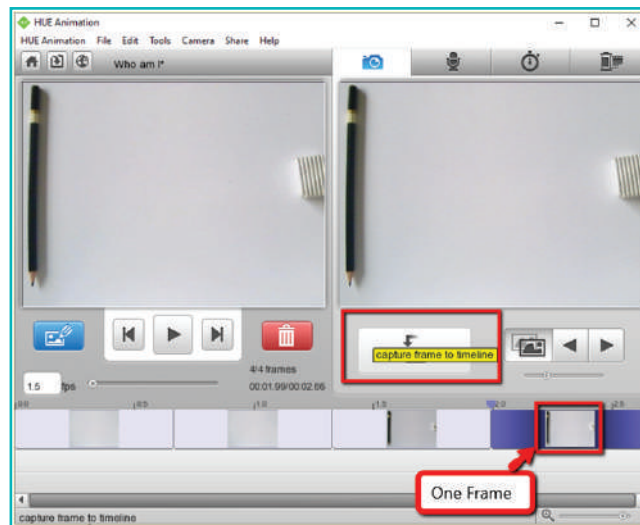


Capture frame button

Steps

To show the pencil and eraser as they meet, we move them closer.

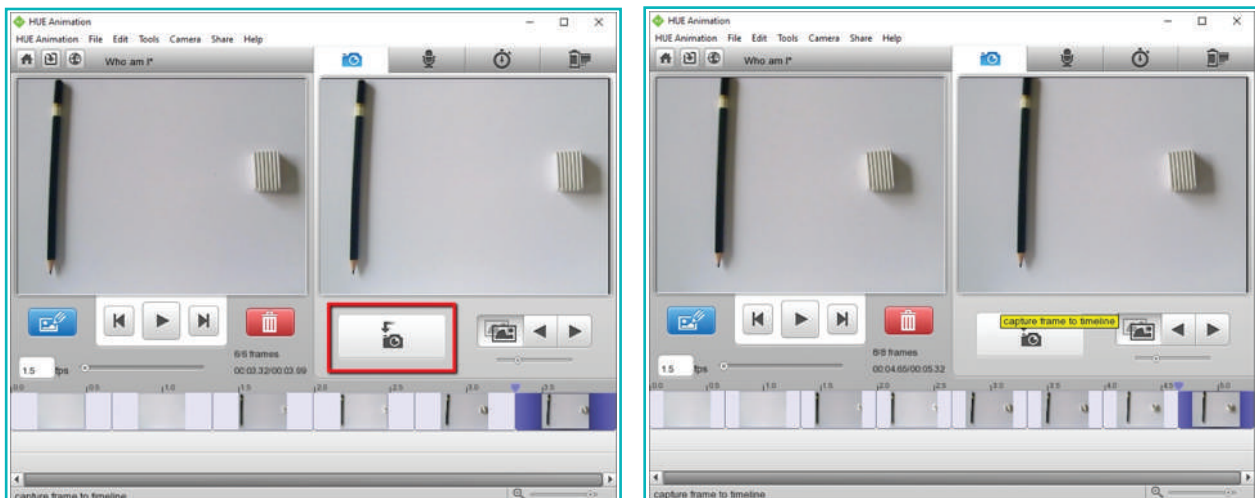
- We want to show the pencil and eraser as they enter the frame. Position the pencil & the eraser on the edges of the paper.
- Capture the same frame two times.



Enter pencil and eraser

A frame is a single image in a sequence of images.

- Move the pencil and eraser towards each other as if they are moving closer.
- Capture the same frame two times.



Capture frames two times

Bytes

We sometimes capture each frame twice. This makes the animation look better. It gives the viewer time to read any text on the screen.

12. Introduction to Animation

Steps



- Use the other sheet of paper to cut out speech bubbles.
- The speech bubbles are to make the objects talk.
- Make five speech bubbles.

- Write this conversation on the speech bubbles. Each line will be in a separate speech bubble.

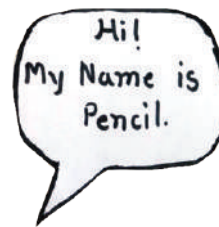
Pencil: Hi, my name is Pencil

Eraser: Hello, I am Eraser

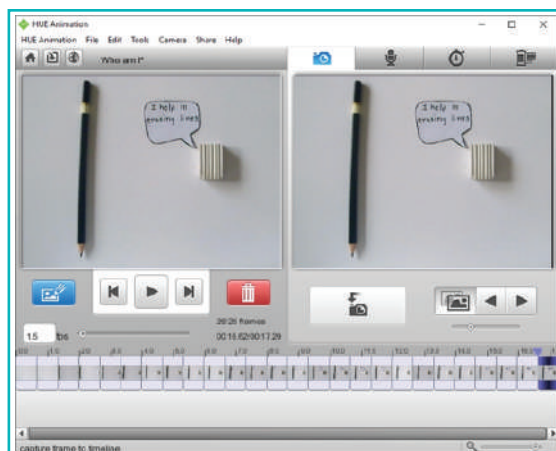
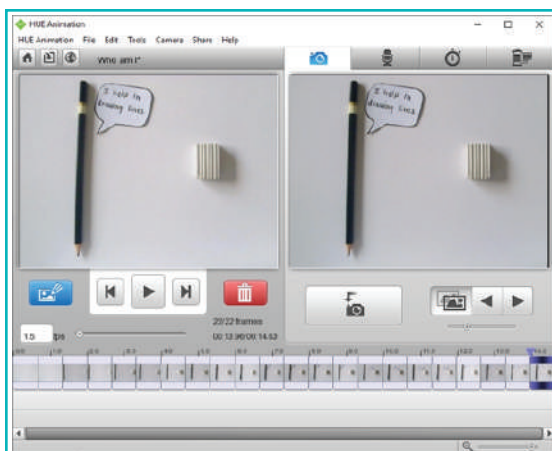
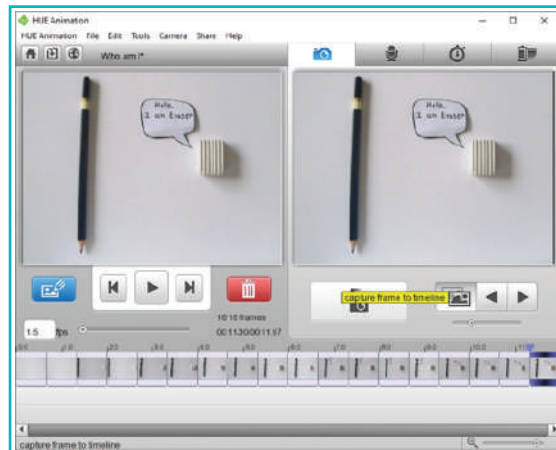
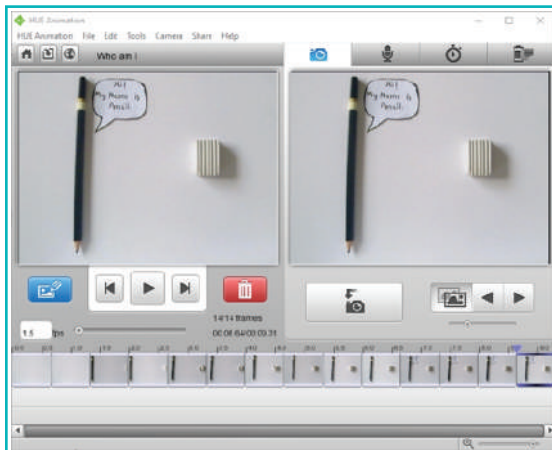
Pencil: I help you to draw lines

Eraser: I help you to erase lines

Both: You can find us in a pencil box!

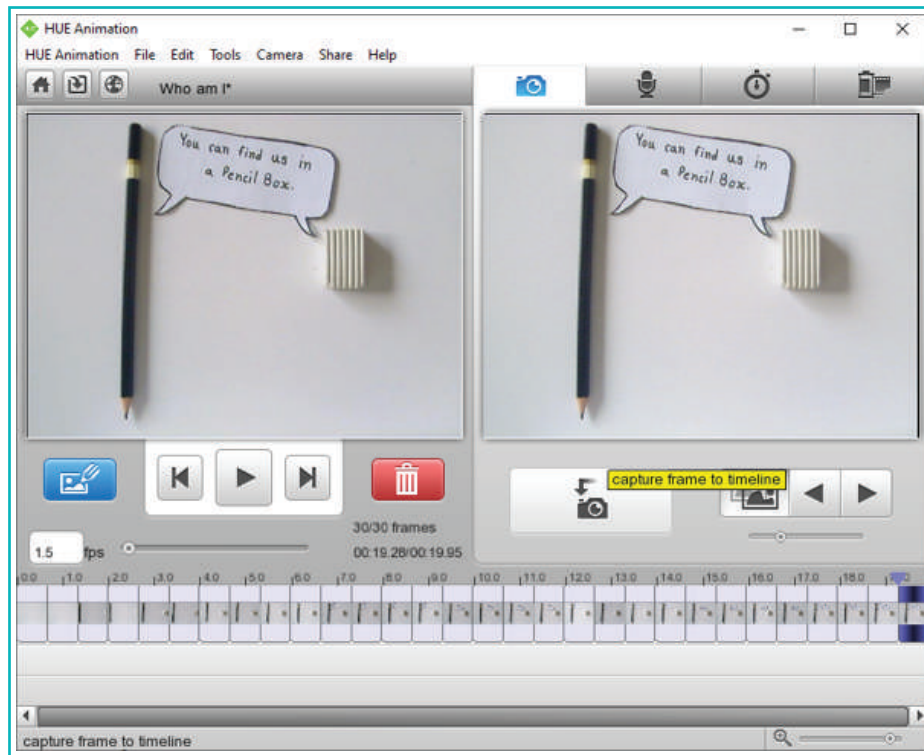


- Place the speech bubbles into the frame one at a time. Pencil introduces itself first, and then Eraser. Follow the images given below.
- Capture each frame two times.



Steps

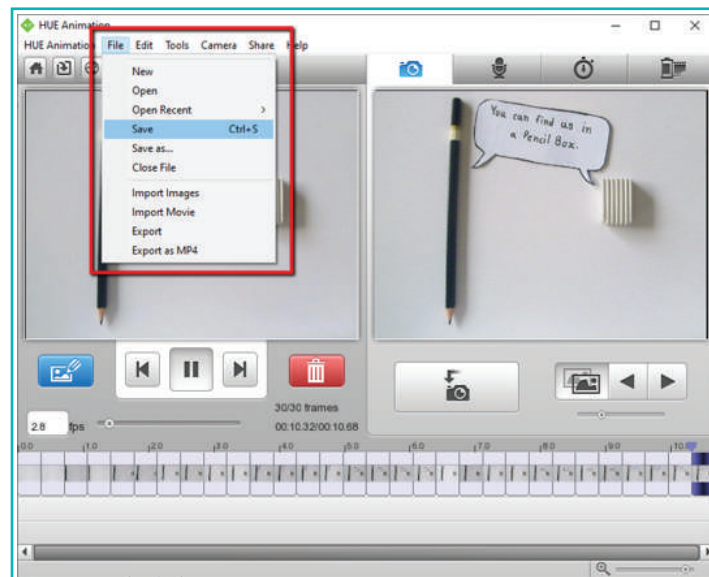
- In the third frame, Pencil and Eraser talk together. Follow the image below. Capture this frame twice.



Introduction frames

5 Save the file

- Click on File and select Save

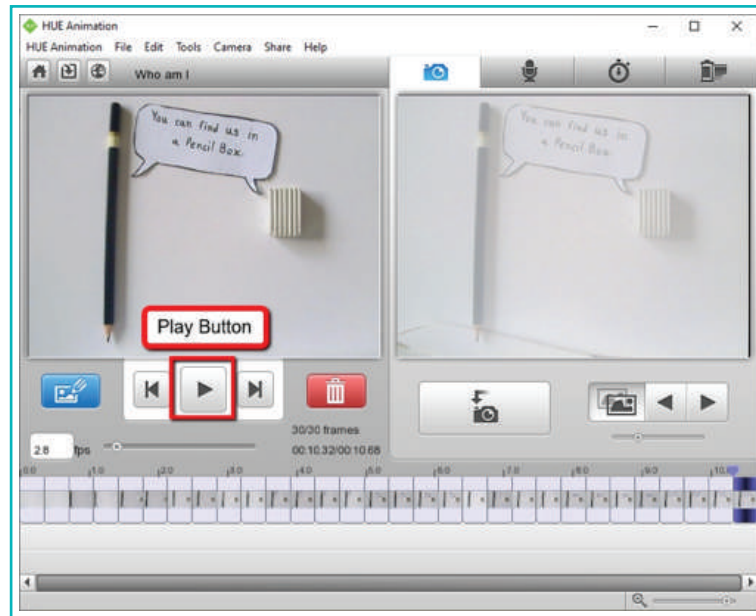


Save the file

Steps

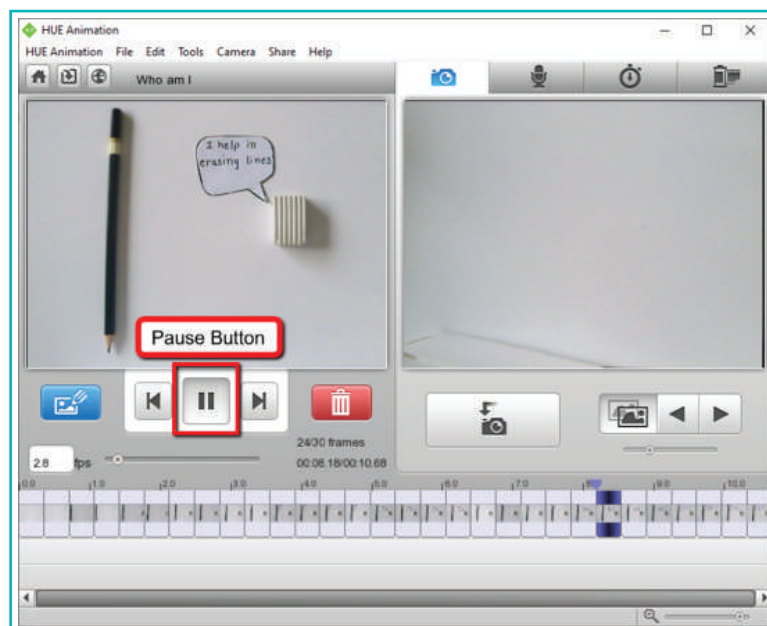
6 Play the animation

- We have captured the images we need for our animation project.
- These separate images will be turned into a movie. For this, we use the Hue Animation software.
- To run this animation, click the **Play** button on the screen.



Play animation

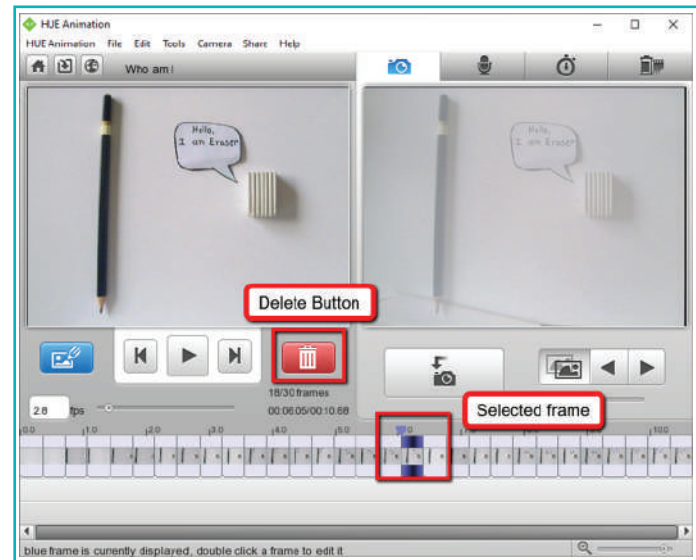
- To pause the animation, click the **Pause** button on the screen.



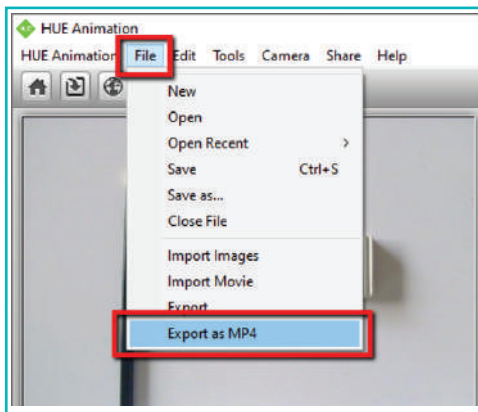
Pause animation

Steps

- To delete a frame, select it and click on the **Delete** button.
- We do not need to delete the entire animation. We can delete only unwanted frames.



Delete Frame



Export as MP4

- 7 Create a video file of animation
 - Create a video file of the animation.
 - Go to File and select '**Export as MP4**'. This project gets saved as a movie in our computer.

Healthy Practices:

Put the animation kit back in the box when not in use

Bytes

A video file can be played anytime on the computer if it is saved. We do not have to open the Hue Animation software each time.

Exercise

 State whether True or False

3 The play button is used to run the animation.

 _____

4 We click on the save button to take a new picture.

 _____

5 We can delete unwanted frames. There is no need to delete the entire animation.

 _____

We can use this to make an animation with any object around us.

Yes, I can't wait to try this with my toys at home! I will make them talk, just like Toy Story!



Tech Flash

- 1. Animation** : Animation is giving movement to drawings, models and non-living objects. It makes them appear as if they have life.
- 2. Capture** : To record a still or moving image or video.
- 3. Camera** : A device that takes pictures or videos.
- 4. Windows** : A separate viewing area on a computer screen.
- 5. Frames** : Individual pictures in a sequence of images.
- 6. Project** : A project is a set of tasks.
- 7. Save** : Stores a file on our computer.
- 8. Button** : Run the animation.
- 9. Pause Button** : Stops the animation for a short period of time.
- 10. Delete Button** : Removes a frame that we do not want.